

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

Probability and conditional probability

02

4 questions · ~6 min

United States
Presidents
from 1789 to 2015

Ages	Number
40–44	2
45–49	7
50–54	13
55–59	11
60–64	7
65–69	3

The table above gives the number of United States presidents from 1789 to 2015 whose age at the time they first took office is within the interval listed. Of those presidents who were at least 50 years old when they first took office, what fraction were at least 60 years old?

- A. $\frac{10}{43}$
- B. $\frac{10}{34}$
- C. $\frac{10}{24}$

D.

$$\frac{25}{34}$$

Coat color	Eye color		
	Deep blue	Light brown	Total
Cream-tortoiseshell	16	16	32
Chocolate	12	4	16
Total	28	20	48

The data on the coat color and eye color for 48 Himalayan kittens available for adoption were collected and summarized in the table above. What fraction of the chocolate-colored kittens has deep blue eyes?

- A. $\frac{12}{48}$
- B. $\frac{12}{28}$
- C. $\frac{16}{32}$
- D. $\frac{12}{16}$

The table summarizes the distribution of color and shape for 100 tiles of equal area.

	Red	Blue	Yellow	Total
Square	10	20	25	55
Pentagon	20	10	15	45
Total	30	30	40	100

If one of these tiles is selected at random, what is the probability of selecting a red tile? (Express your answer as a decimal or fraction, not as a percent.)

STUDENT-PRODUCED RESPONSE

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Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

	Human Resources	Accounting
Bachelor's degree	4	3
Master's degree	2	6

The table above shows the number of people who work in the Human Resources and Accounting departments of a company and the highest level of education they have completed. A person from one of these departments is to be chosen at random. If the person chosen works in the Human Resources department, what is the probability that the highest level of education the person completed is a master's degree?

- A. $\frac{2}{15}$
- B. $\frac{1}{3}$
- C. $\frac{1}{4}$
- D. $\frac{8}{15}$